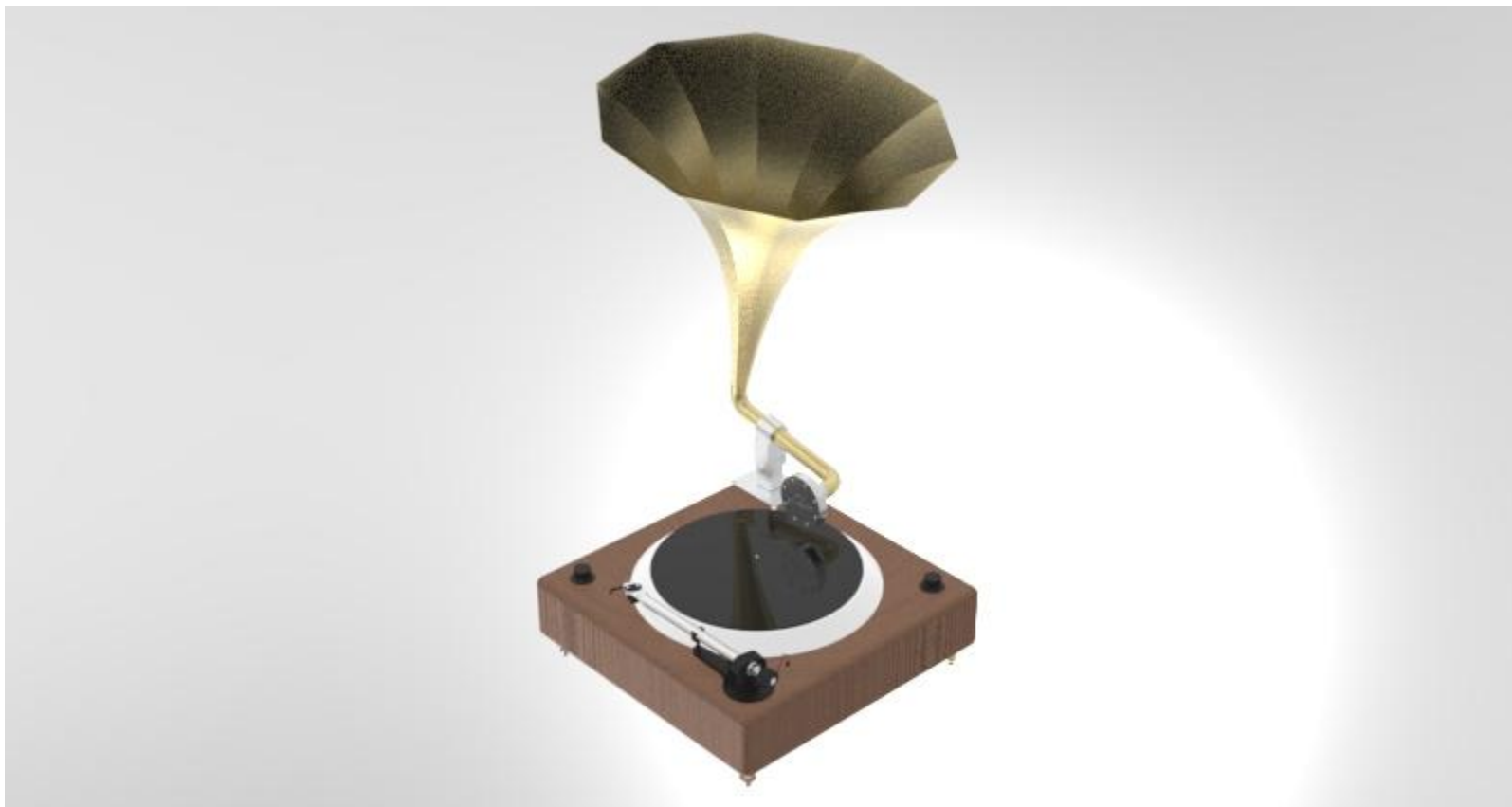


5010 Final CDR

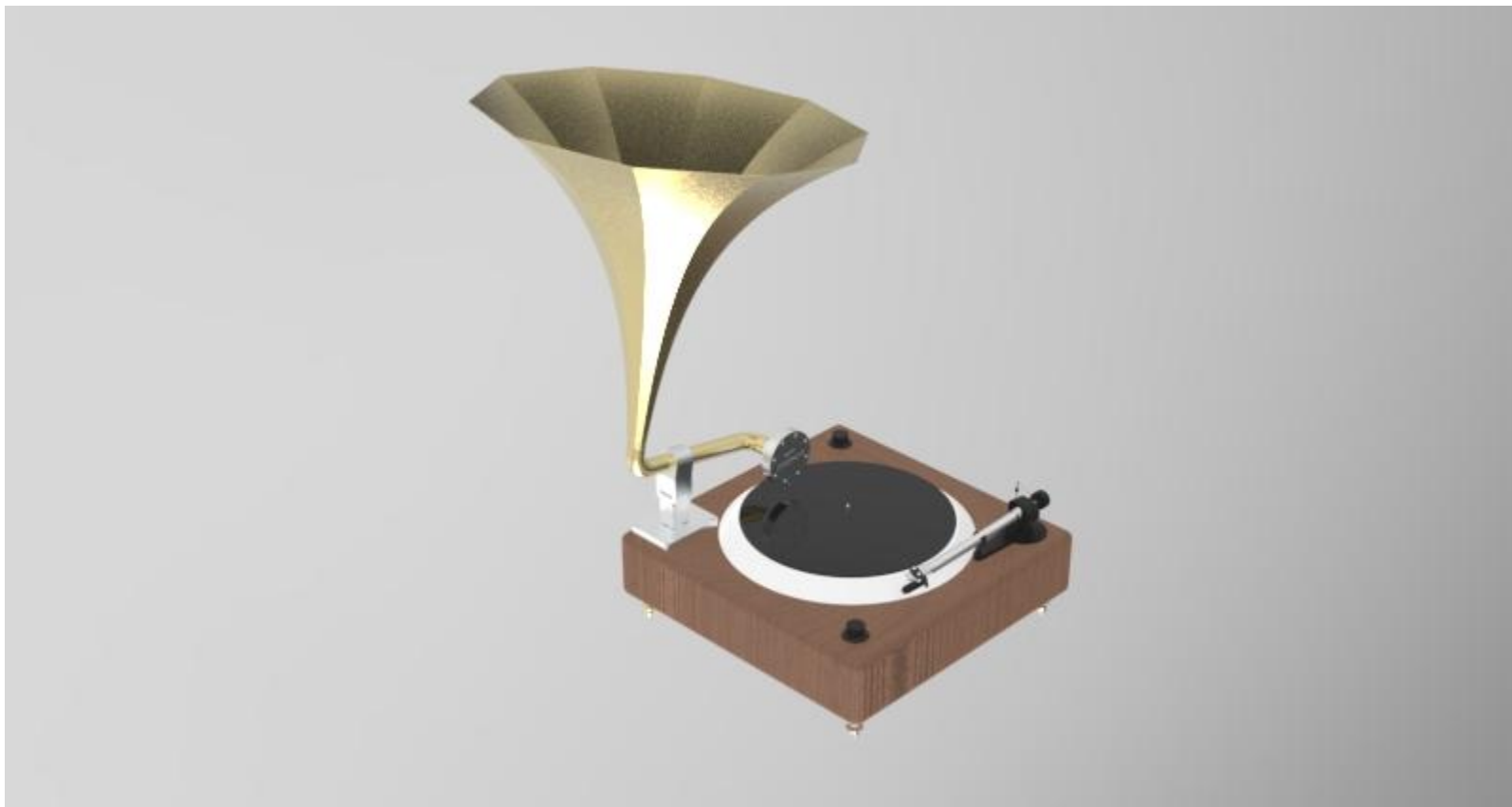


GraMAFFone by CNC Symphony

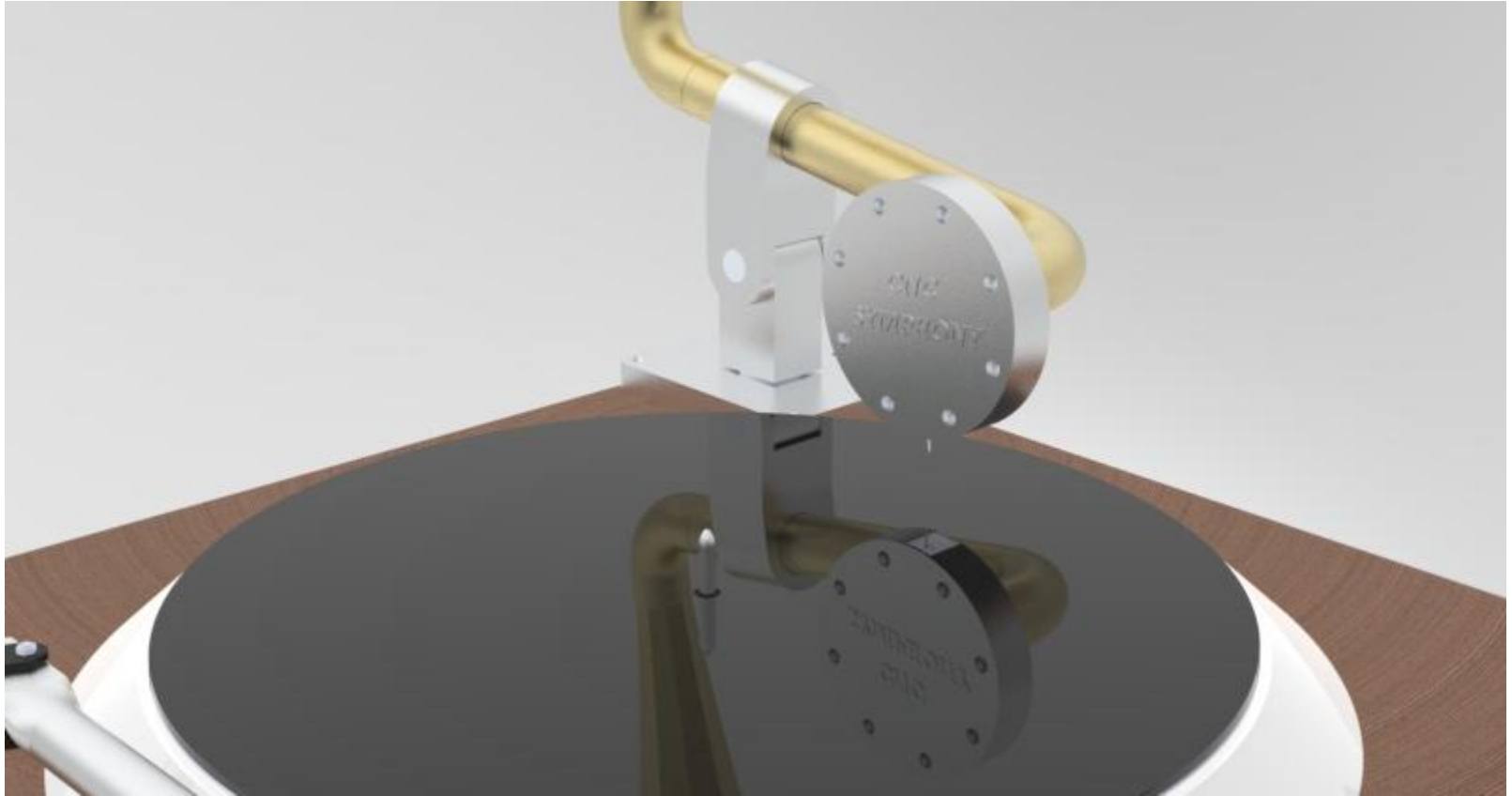
Final CAD Renderings



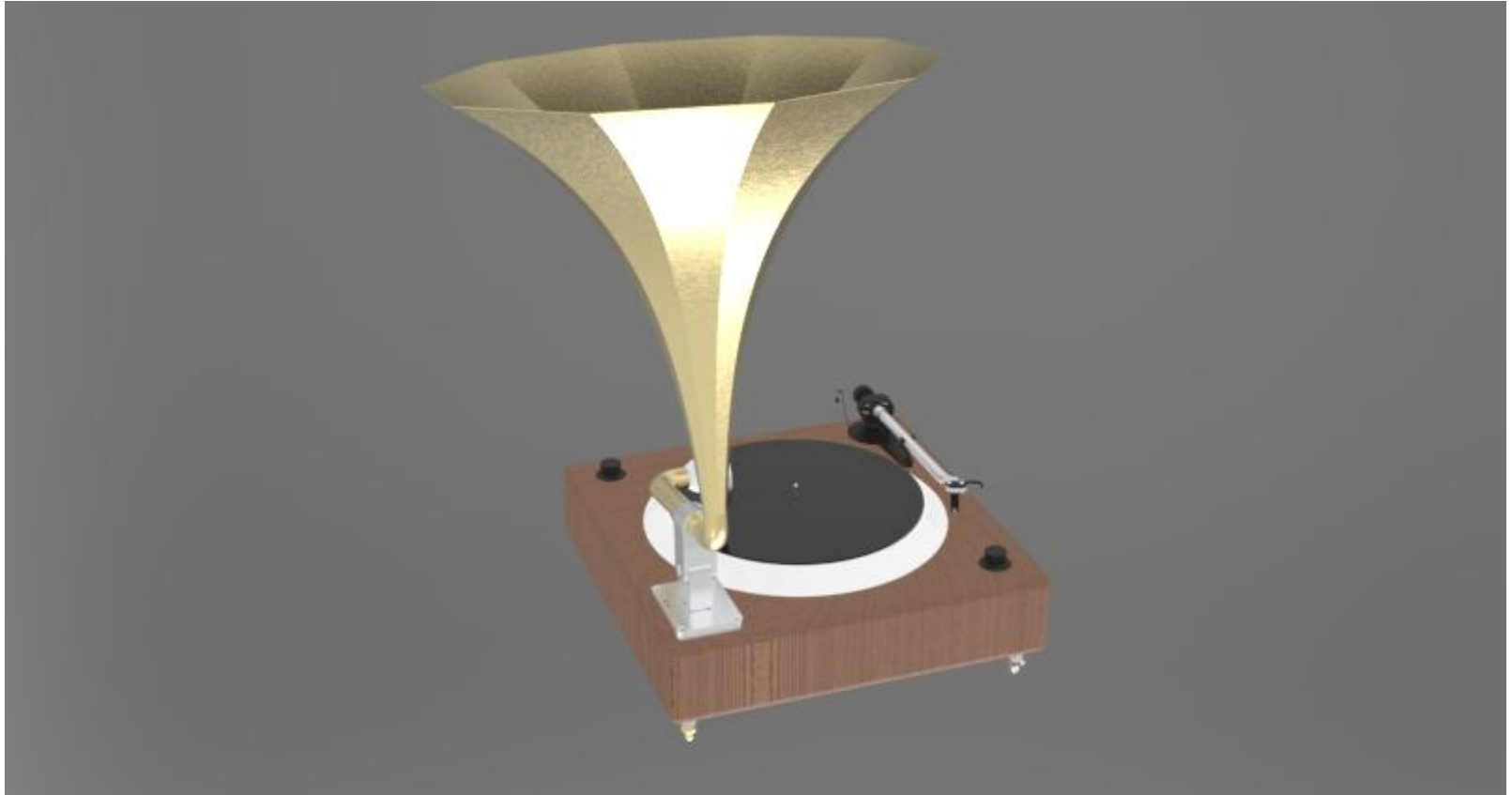
Final CAD Renderings



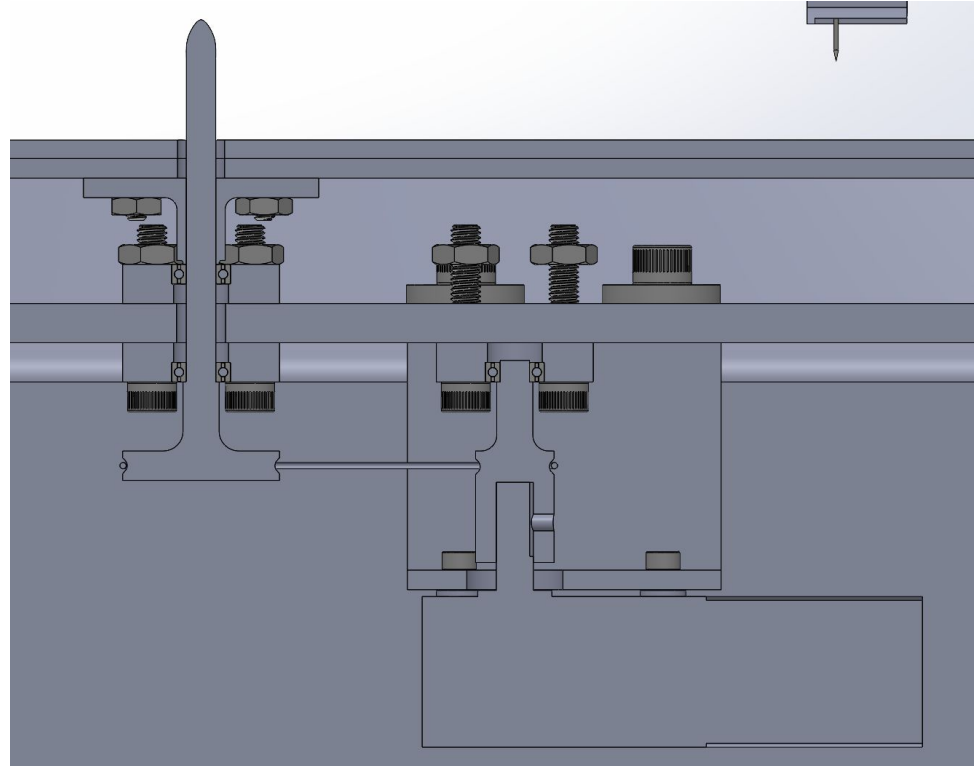
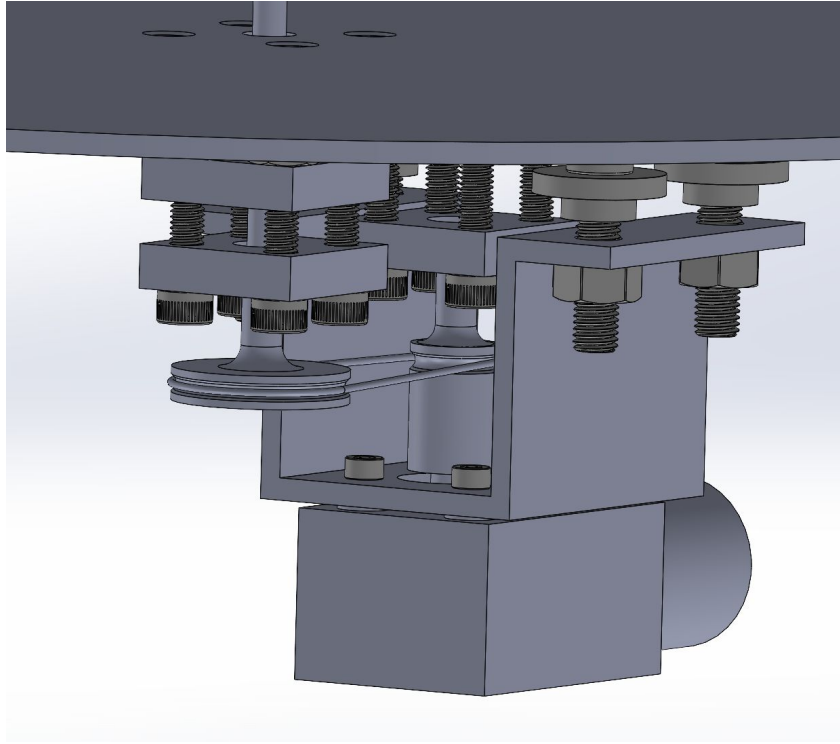
Final CAD Renderings



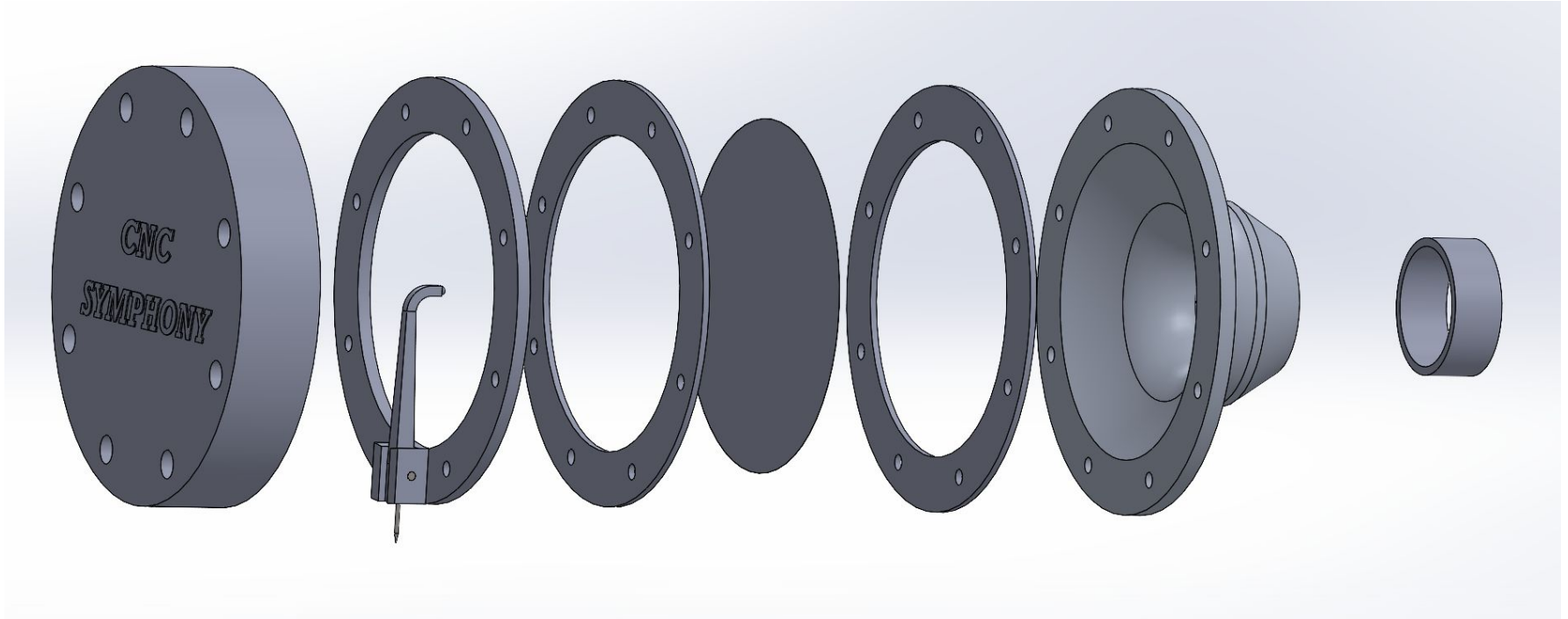
Final CAD Renderings



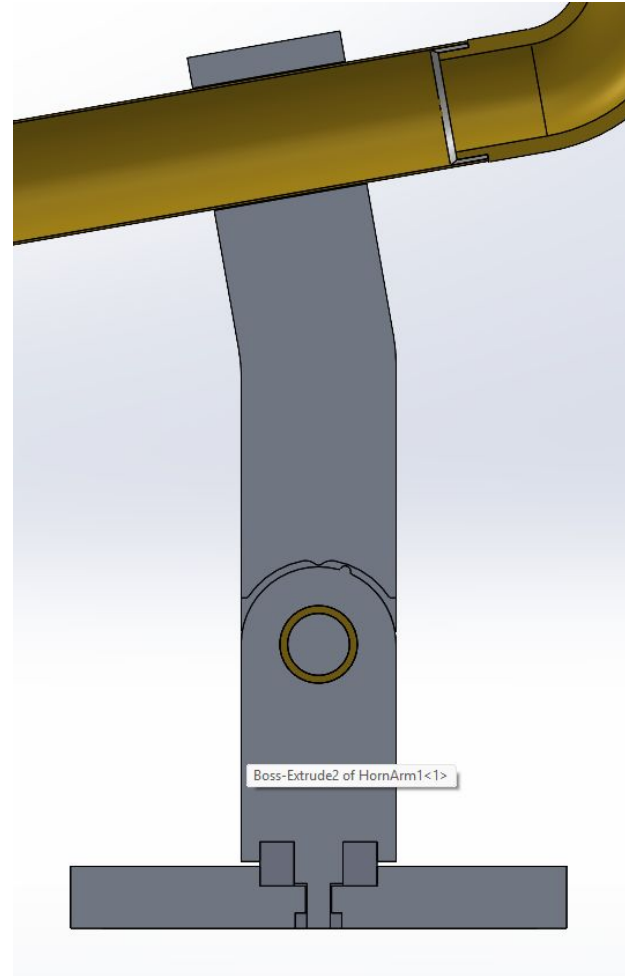
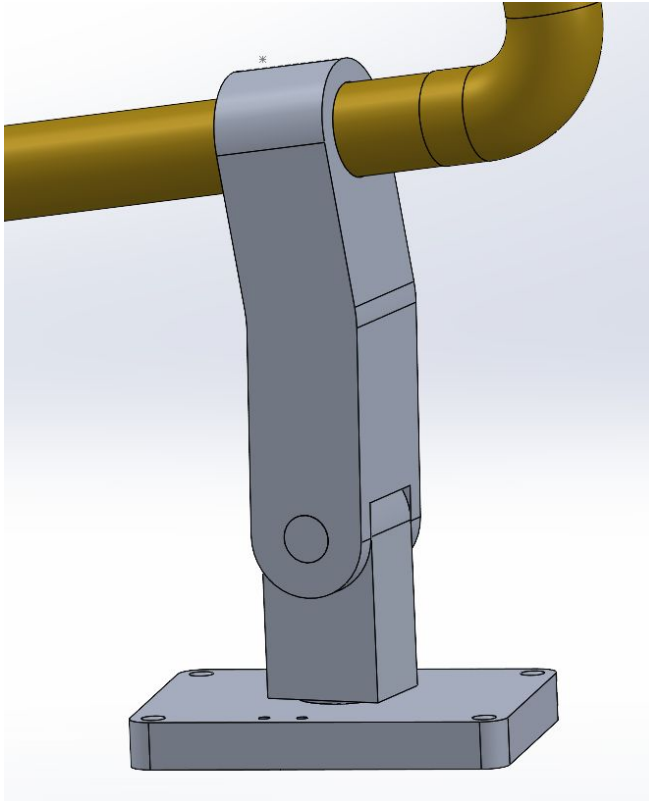
Progress Update: Motor Subassembly



Progress Update: Diaphragm & Sound Box CAD



Progress Update: Pivot CAD

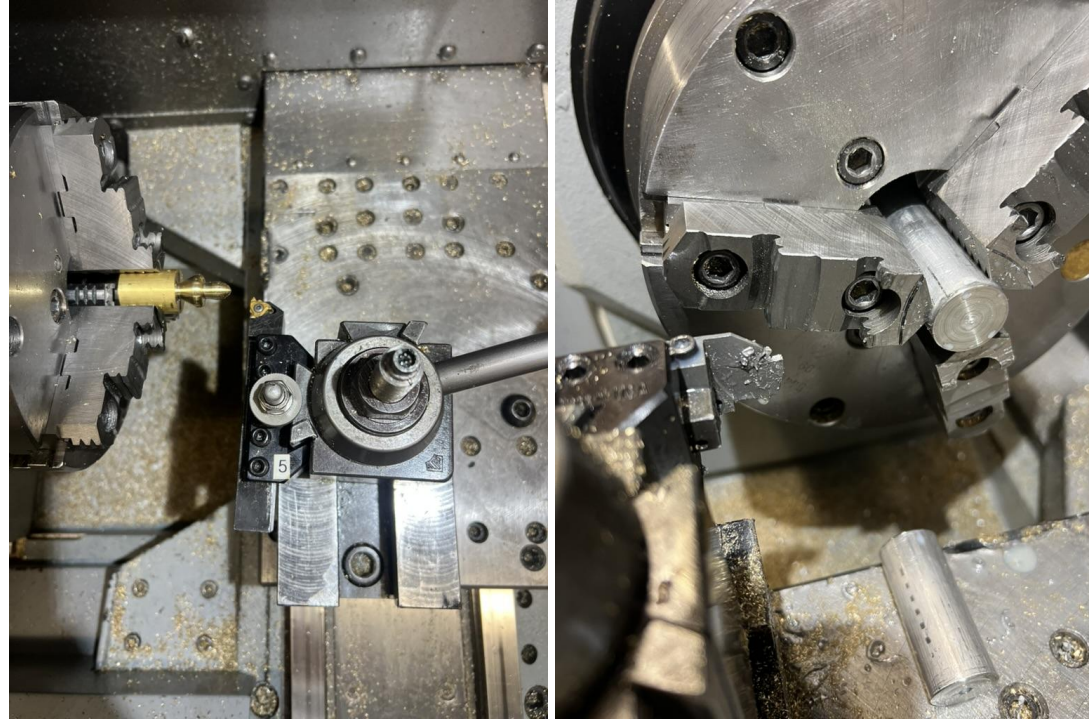


Progress Update: Machining

Not a good day on the TL-1 yesterday:(

- Tool holder sideways (fixed now)
- Part-off tool different?
- 2 “vintage” legs done
- 2 “modern” legs will be finished right after this lecture

3D printed prototype diaphragm box



Final Schedule

November 10 - 15:

- Note: CAD Freeze 11/12
- Submit full BOM for orders 11/10
- Machining: Feet, Pivot, Pivot Plate, Motor Mount Bracket, Motor Shaft
- Make prototype horn for testing components
- Test 3D print sound box & iterate on design

November 16 - 22:

- Machining: Sound Box, Tone Arm, Electronic Housings & Mounts
- Injection Molding: Knob covers
- Garage Lab: make wooden/acrylic base

November 23 - 29: THANKSGIVING

December 1 - 6:

- ***HORN: Form & Solder (NOTE: do this as soon as the order comes in...putting later assuming it will take a while)
- Assembly
- Testing

December 7 - 10:

- Assembly
- Finishes
- Testing

December 11: DEMO DAY

Final BOM & Budget

Part	Description	Size	Unit Cost	Quantity	Total Cost	Link
Wood	Base	1/2" x 6" x 48"	\$18.53	2	\$37.06	https://www.homedepot.com/p/Weaber-1-2-in-x-6-in-x-48-in-Wood-Base-for-Turntables-1-Piece-Pack-of-2/31911111
Acrylic	Base	18" x 24" x 3/32"	\$18.48	2	\$36.96	https://www.homedepot.com/p/OPTIX-18-in-x-24-in-Acrylic-Base-for-Turntables-1-Piece-Pack-of-2/31911111
Electric Tonearm	Electric half	N/A	\$15.19	1	\$15.19	https://a.co/d/8TxOpBg
Brass Horn	Outsource	N/A	\$540.00	0	\$0.00	Oshcut
Brass Horn	Outsource	N/A	\$443.00	1	\$443.00	Send cut send
Aluminum Round Stock	stock for TL-1 machined pieces	d = 1", L = 24"	\$18.09	0	\$0.00	https://www.mcmaster.com/products/metals/aluminum-round-stock/
Tone arm stock	Adam to go buy		\$4.00	1	\$4.00	
Brass Round Stock	stock for TL-1 machined pieces	d = 1", L = 24"	\$35.65	0	\$0.00	https://www.mcmaster.com/products/metals/brass-round-stock/
Aluminum Square Stock	stock for Minimill machined pieces	2" x 24" x 2"	\$153.43	1	\$153.43	https://www.mcmaster.com/products/metals/aluminum-square-stock/
Brass Square Stock	stock for Minimill machined pieces	2"x2"x6"	\$157.89	0	\$0.00	https://www.mcmaster.com/products/brass-square-stock/
Needle	For mechanical side, splitting with phonograph team	N/A	\$19.97	0.5	\$9.99	https://a.co/d/itqc3IA
Motor	for turntable	12V, 120rpm	\$15.99	1	\$15.99	https://www.amazon.com/CHANCS-Geared-Nesting-Motor/dp/B000TQXQ2/ref=sr_1_1?dref=ad
Battery	powers motor & electrical half	3600 mAh, 14.8V	\$26.83	1	\$26.83	https://a.co/d/dof8WA
Speaker	3.5" speakers	80W	\$49.99	1	\$49.99	https://www.amazon.com/dp/B00EDTSXQ2/ref=sr_1_1?dref=ad
isolators for motor mount	40lb cap. isolators	.75"x.26"	\$13.72	1	\$13.72	https://www.mcmaster.com/products/9311k53/
Encoder			\$28.25	1	\$28.25	https://www.digikey.com/en/products/detail/samsongm512/9376K571
Vibration Dampers	for legs		\$2.63	4	\$10.52	9376K571
motor assy bearings	for motor assy and pivot		\$9.49	1	\$9.49	https://www.amazon.com/uxcell-16-inchx3-8-inch-Diameter-Ball-Bearings/dp/B000TQXQ2/ref=sr_1_1?dref=ad
Thrust bearings	for pivot mech		\$7.69	1	\$7.69	https://www.amazon.com/uxcell-Single-Direction-Thrust-Bearing/dp/B000TQXQ2/ref=sr_1_1?dref=ad
Ball bearings	for pivot mech		\$7.79		\$7.79	uxcell 6700ZZ Deep Groove Ball Bearings Z2 10mm ID X 16mm OD X 5mm WIDTH
Big Elbow			\$45.00	0	\$0.00	https://a.co/d/4Ul6dbc
Elbow	Adam get in LA		\$25.00	2	\$50.00	https://tdibrass.com/products/ff122b-flush-curve-elbow
TDA7297	audio amp board		\$10.49	1	\$10.49	https://www.amazon.com/HiLetgo-TDA7297-Versatile-Audio-Amplifier-Board/dp/B000TQXQ2/ref=sr_1_1?dref=ad
L298 Driver	L298 motor driver		\$9.99	1	\$9.99	https://www.amazon.com/BOJACK-H-Bridge-Controller/dp/B000TQXQ2/ref=sr_1_1?dref=ad
TOTAL					\$940.38	